

NAKHON PHANOM AGROMET, Thailand

WMO: 483580

Lat: 17.412N

Lon: 104.779E

Elev: 153

StdP: 99.50

Time Zone: 7.00 (E07)

Period: 08-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 11.4	(c) 13.3	(d) 7.3	(e) 6.4	(f) 16.3	(g) 8.8	(h) 7.2	(i) 16.3	(j) 7.9	(k) 24.3	(l) 6.4	(m) 23.3	(n) 0.1	(o) N/A	(p) 0.288

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 4	(b) 10.3	(c) 37.1	(d) 25.6	(e) 35.7	(f) 25.7	(g) 34.5	(h) 25.7	(i) 28.1	(j) 32.9	(k) 27.6	(l) 32.2	(m) 27.2	(n) 31.6	(o) 1.3	(p) N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
26.9	23.0	30.5	26.5	22.5	30.0	26.2	22.0	29.5	91.3	33.1	89.3	32.4	87.5	31.8	34.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years						
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max				
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
5.5	4.0	3.2	7.7	38.5	2.2	1.1	6.1	39.2	4.8	39.9	3.6	40.5	1.9	41.2		
			7.3	30.1	2.2	2.1	5.6	31.6	4.3	32.8	3.1	34.0	1.4	35.5		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	26.4	22.0	24.2	27.3	29.1	28.9	28.5	27.6	27.7	27.7	26.8	25.2	22.3
	DBStd	3.24	2.84	3.27	2.98	2.31	1.73	1.52	1.62	1.35	1.29	1.50	2.22	2.92
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD18.3	12	6	2	1	0	0	0	0	0	0	0	0	4
	CDD10.0	6003	372	397	536	573	586	555	546	548	532	522	455	383
	CDD18.3	2974	118	166	278	323	327	305	287	289	282	263	206	129
	CDH23.3	30045	968	1559	2957	3771	3681	3378	2821	2893	2792	2396	1772	1058
(13)	CDH26.7	11872	307	697	1438	1904	1593	1286	872	884	937	915	680	359
(14) Wind	WSAvg	0.7	1.0	1.0	0.9	0.7	0.5	0.4	0.4	0.4	0.4	0.5	0.8	1.0
(15) Precipitation	PrecAvg	2422	2	12	51	89	259	564	432	660	296	62	5	2
	PrecMax	2837	10	49	118	204	443	1033	771	1233	587	291	45	29
	PrecMin	1820	0	0	6	9	164	288	271	274	97	7	0	0
	PrecStd	347	3	15	37	56	69	225	125	263	133	69	11	7
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	32.6	35.8	38.1	39.4	37.3	35.6	34.7	34.0	34.2	34.1	33.9	33.1
		MCWB	22.7	23.9	24.7	25.7	26.6	27.7	27.1	27.4	26.8	25.6	25.2	23.4
	2%	DB	31.2	34.4	36.5	37.7	35.7	34.2	33.0	32.6	33.1	33.0	32.7	31.8
		MCWB	22.0	23.5	24.9	25.9	27.0	27.1	27.1	26.9	26.5	25.0	24.7	22.8
	5%	DB	30.0	32.9	35.0	35.9	34.3	32.9	31.8	31.7	32.2	32.0	31.7	30.3
		MCWB	21.5	23.1	24.7	25.7	26.7	27.0	26.7	26.8	26.5	24.8	24.0	22.3
	10%	DB	28.4	31.2	33.4	34.3	32.9	31.9	30.6	30.7	31.2	31.1	30.5	28.7
		MCWB	21.0	22.5	24.2	25.6	26.4	26.7	26.6	26.5	26.3	24.6	23.5	21.4
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	25.1	25.7	26.6	28.2	28.5	28.7	28.3	28.2	28.1	27.1	26.4	24.7
		MCDB	29.2	33.1	34.1	34.7	34.4	33.5	32.5	32.1	32.2	32.1	31.5	30.8
	2%	WB	23.4	24.9	25.8	27.2	27.8	27.9	27.5	27.6	27.4	26.2	25.5	23.7
		MCDB	28.6	32.0	33.5	34.2	33.5	32.2	31.3	31.3	31.5	30.8	31.0	30.1
	5%	WB	22.5	24.1	25.3	26.7	27.2	27.5	27.1	27.1	27.0	25.7	24.9	22.8
		MCDB	28.0	31.0	32.7	33.3	32.5	31.5	30.6	30.7	30.9	29.9	30.0	29.1
	10%	WB	21.5	23.2	24.8	26.1	26.8	27.0	26.7	26.7	26.6	25.2	24.2	21.9
		MCDB	27.2	29.2	31.7	32.1	31.5	30.9	29.8	29.9	30.1	29.1	28.8	27.6
(35) Mean Daily Temperature Range	5% DB	MDBR	12.5	12.7	11.2	10.3	8.0	6.4	5.4	5.6	6.5	8.7	10.8	12.5
		MCDBR	13.4	14.2	12.8	12.2	9.9	8.0	7.4	7.3	8.2	9.8	11.4	13.0
		MCWBR	5.4	5.0	3.9	3.7	3.3	2.6	2.7	2.7	3.0	3.3	4.4	5.2
	5% WB	MCDBR	12.0	12.9	11.7	10.6	8.7	7.4	6.8	6.8	7.6	8.4	10.4	12.2
		MCWBR	5.6	5.3	4.1	3.9	3.3	3.0	2.9	2.9	3.1	3.2	4.6	5.3
(40) Clear-Sky Solar Irradiance	taub		0.452	0.558	0.662	0.632	0.512	0.474	0.462	0.474	0.504	0.512	0.454	0.455
	taud		2.174	1.895	1.695	1.801	2.154	2.282	2.313	2.266	2.171	2.116	2.251	2.186
	Ebn at Noon		828	756	693	718	800	822	834	830	806	786	821	811
	Edh at Noon		142	196	247	223	155	135	131	138	151	155	130	137
(44) All-Sky Solar Radiation	RadAvg		4.67	5.00	4.98	5.33	5.09	4.66	4.30	4.30	4.52	4.73	4.65	4.41
	RadStd		0.28	0.36	0.21	0.29	0.40	0.45	0.53	0.37	0.40	0.41	0.21	0.17

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (10 neighbors)	+0.30	N/A	N/A	N/A	N/A	+0.20	+0.15	N/A	N/A	N/A

Nomenclature: See separate page